

Curriculum Vitae — Backend Software Engineer & Systems Architect

EXECUTIVE PROFILE

DETAILED TECHNICAL SKILLS MATRIX

- ## COMPREHENSIVE PROJECT ARCHIVE & CONTRIBUTIONS

- Architected a zero-framework native Node.js REST application server relying solely on core standard library modules (node:http, node:fs).

- Implemented an NP-hard Constraint Satisfaction Problem (CSP) Backtracking algorithm that resolves complex faculty-venue constraints with zero collisions.
- Integrated K-Means dynamic clustering and Bayesian statistical classification for anti-cheating exam seating distribution.

5. NVIDIA RTX-LocalAI Enterprise Execution Suite

Specialization: C++20, CUDA 12.6, Memory Management, Speculative Decoding | github.com/hrlpavan/nvidia-project-by-hrl

- Engineered an on-device AI inference engine with virtual-to-physical Paged KV-Cache memory allocator (0.17 μ s latency, 0.00% fragmentation).
 - Implemented CUDA Graph static capture and replay, slashing host-to-device dispatch overhead to under 5 μ s.
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PROFESSIONAL EXPERIENCE & ENTREPRENEURSHIP

Founder & Managing Director / Chief Systems Architect

HRL International Private Limited | 2026 – Present

- Lead architectural design, code review protocols, and technical governance across 31+ open-source and proprietary software systems.
 - Secured corporate innovation grant allocations under the Startup India Seed Fund Scheme (SISFS) and Karnataka ELEVATE 2026 initiative.
 - Published the master engineering treatise 'Rule Breaking (Volume 4.0): The 100x AI Chief Architect Manifesto'.
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EDUCATION & PROFESSIONAL CREDENTIALS

Bachelor of Engineering (B.E.) in Computer Science & Engineering (AI & ML)

Sahyadri College of Engineering & Management (SCEM), Mangaluru, Karnataka, India

- Academic Honors: Advanced coursework in Systems Architecture, Operating Systems, Relational Databases, and Artificial Intelligence.
- **JPMorgan Chase & Co.:** Distributed Streaming & Software Engineering Simulation (Kafka Streaming, REST Microservice Design).
- **Deloitte Australia:** Enterprise Cybersecurity Defense & Threat Assessment.